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Să se arate că dacă f este convexă pe $[0, 1]$ atunci și $g(x) = \sum_{k=0}^m b_{m,k}(x) \int_0^1 b_{m,k}(t) f(t) dt$ este convexă pe $[0, 1]$.

f -convexă pe $[0, 1] \Rightarrow \forall a \in [0, 1]$ și $x_1, x_2 \in [0, 1]$
 $f((1-a)x_1 + ax_2) \leq (1-a)f(x_1) + af(x_2)$

g "diferă puțin" de "polinomul Bernstein"

$$(B_m f)(x) = \sum_{k=0}^m b_{m,k}(x) \int_0^1 b_{m,k}(t) f(t) dt$$

g -polinom de grad " m " $\Rightarrow g \in C^\infty[0, 1]$

$$g'(x) = \left(\sum_{k=0}^m \binom{m}{k} x^k (1-x)^{m-k} \int_0^1 b_{m,k}(t) f(t) dt \right)' =$$

$$= \sum_{k=0}^m \left(\binom{m}{k} k x^{k-1} (1-x)^{m-k} - \binom{m}{k} (m-k) x^k (1-x)^{m-k-1} \right) \cdot \int_0^1 b_{m,k}(t) f(t) dt =$$

$$\cdot \int_0^1 b_{m,k}(t) f(t) dt =$$

$$= \sum_{k=1}^m \binom{m}{k} \cdot k x^{k-1} (1-x)^{m-k} \int_0^1 b_{m,k}(t) f(t) dt -$$

$$- \sum_{k=0}^{m-1} \binom{m}{k} (m-k) x^k (1-x)^{m-k-1} \int_0^1 b_{m,k}(t) f(t) dt =$$

$$= m \sum_{k=1}^m \binom{m-1}{k-1} x^{k-1} (1-x)^{(m-1)-(k-1)} \int_0^1 b_{m,k}(t) f(t) dt -$$

$$- m \sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{(m-1)-k} \int_0^1 b_{m,k}(t) f(t) dt$$

$$= m \sum_{i=0}^{m-1} \binom{m-1}{i} x^i (1-x)^{(m-1)-i} \int_0^1 b_{m,i+1}(t) f(t) dt -$$

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$$= m \sum_{k=0}^m \binom{m-1}{k} x^k (1-x)^{(m-1)-k} - h \int_0^1 b_{m,h}(t) f(t) dt =$$

$$= m \sum_{i=0}^{m-1} b_{m-1,i}(x) \int_0^1 b_{m,i,i}(t) f(t) dt - m \sum_{k=0}^{m-1} b_{m-1,k}(x) \int_0^1 b_{m,k}(t) f(t) dt$$

$$= m \sum_{k=0}^{m-1} b_{m-1,k}(x) \int_0^1 (b_{m,k+1}(t) f(t) - b_{m,k}(t) f(t)) dt =$$

$$= m \sum_{k=0}^{m-1} b_{m-1,k}(x) \int_0^1 \frac{-b'_{m+1,k+1}(t)}{m+1} f(t) dt =$$

$$\uparrow$$

$$b'_{m,k}(x) = m(b_{m-1,k-1}(x) - b_{m-1,k}(x))$$

$$= \frac{m}{m+1} \sum_{k=0}^{m-1} b_{m-1,k}(x) \left[-b_{m+1,k+1}(t) f(t) \right]_{t=0}^{t=1} + \int_0^1 b_{m+1,k+1}(t) f'(t) dt$$

= 0

$$= \frac{m}{m+1} \sum_{k=0}^{m-1} b_{m-1,k}(x) \int_0^1 b_{m+1,k+1}(t) f'(t) dt$$

$$\Rightarrow g'(x) = \frac{m}{m+1} \sum_{k=0}^{m-1} b_{m-1,k}(x) \int_0^1 b_{m+1,k+1}(t) f'(t) dt$$

$$g''(x) = \frac{m}{m+1} \left(\sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{(m-1)-k} \right) \int_0^1 b_{m+1,k+1}(t) f'(t) dt$$

$$= \frac{m}{m+1} \left(\sum_{k=0}^{m-1} \binom{m-1}{k} k x^{k-1} (1-x)^{(m-1)-k} - \sum_{k=0}^{m-1} \binom{m-1}{k} x^k (1-x)^{(m-1)-k} \right)$$

$$\cdot (1-x)^{(m-1)-k-1} \int_0^1 b_{m+1,k+1}(t) f'(t) dt =$$

(i=k-1)

$$= \frac{m}{m+1} \left((m-1) \sum_{k=1}^{m-1} \binom{m-2}{k-1} x^{k-1} (1-x)^{(m-2)-(k-1)} - \sum_{k=0}^{m-2} \binom{m-2}{k} x^k (1-x)^{(m-2)-k} \right) \int_0^1 b_{m+1,k+1}(t) f'(t) dt$$

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$$= \frac{n(n-1)}{n+1} \sum_{i=0}^{n-2} b_{n-2,i}(x) \int_0^1 b_{n+1,i+2}(t) f'(t) dt -$$

$$- \frac{n(n+1)}{n+1} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 b_{n+1,k+1}(t) f'(t) dt =$$

$$= \frac{n(n-1)}{n+1} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 (b_{n+1,k+2}(t) - b_{n+1,k+1}(t)) f'(t) dt$$

$$= \frac{n(n-1)}{(n+1)} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 -\frac{b'_{n+1,k+2}(t)}{n+2} f'(t) dt$$

$$= \frac{n(n-1)}{(n+1)(n+2)} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 b_{n+2,k+2}(t) f''(t) dt$$

$$\Rightarrow g''(x) = \frac{n(n-1)}{(n+1)(n+2)} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 b_{n+2,k+2}(t) f''(t) dt$$

In concluzie pt $f \in C^2[0,1]$ avem

$$g'(x) = \frac{n}{(n+1)} \sum_{k=0}^{n-1} b_{n-1,k}(x) \int_0^1 b_{n+1,k+1}(t) f'(t) dt$$

$$g''(x) = \frac{n(n-1)}{(n+1)(n+2)} \sum_{k=0}^{n-2} b_{n-2,k}(x) \int_0^1 b_{n+2,k+2}(t) f''(t) dt$$

f -convexă $\Rightarrow f'' \geq 0 \Rightarrow g'' \geq 0 \Rightarrow g$ convexă

Obs!! Dacă f nu este suficient diferentiată poate fi aproximată uniform continuu cu şirul de polinoame Bernstein asociat lui f